COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	**SECTION:**	CPE21S4
DATE PERFORMED:	09/16/26	**DATE SUBMITTED:**	09/16/26
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SEATWORK NO. 9.1 Implementing Trees

1. PROCEDURE OUTPUT

What procedures did you perform?

1. Code Snippet

Submit the complete code for your BinaryTree class and its main.cpp file.
The code must demonstrate the creation and basic manipulation of a Binary Tree.

2. Documentation
Submit a document explaining the methods/functions of the BinaryTree class you created.
Use your Activity Template 2026

The BinaryTree class is used to create the nodes of a binary tree then the constructor sets the value of the node and makes the left and right child empty at first.



The *insertChildLeft()* function adds a new node on the left side. If there is no left child yet, it just adds the new node. If there is already a left child, the new node will be added first and the old child will be moved under it then the *insertChildRight()* function does the same thing but on the right side. It adds the new node if the right side is empty or moves the existing right child under the new node if there is already one.

```
//insert child:
template<typename T>
void BinaryTree<T>::insertChildLeft(T newNode){
    if(leftChild == nullptr){
        leftChild = new BinaryTree<T>(newNode);
    }else{
        BinaryTree<T> * newChild = new BinaryTree<T>(newNode);

        //move the current left child below the new node
        newChild->leftChild = leftChild;
        leftChild = newChild;
    }
}

template<typename T>
void BinaryTree<T>::insertChildRight(T newNode){
    if(rightChild == nullptr){
        rightChild = new BinaryTree<T>(newNode);
    }else{
        BinaryTree<T> *newChild = new BinaryTree<T>(newNode);
        newChild->rightChild = rightChild;
        rightChild = newChild;
    }
}
```

The *getLeftChild()* and *getRightChild()* functions are used to get the left and right child then the *getRootVal()* gets the value of the current node then the *setRootVal()* is used to change the value of the node.

```
template<typename T>
BinaryTree<T> *BinaryTree<T>::getLeftChild(){
    return leftChild;
}

template<typename T>
BinaryTree<T> *BinaryTree<T>::getRightChild(){
    return rightChild;
}

template <typename T>
T BinaryTree<T>::getRootVal(){
    return key;
}


template <typename T>
void BinaryTree<T>::setRootVal(T newNode){
    key = newNode;
}


#endif
```

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2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Quillbot**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Grammar correction and summarization**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **AI tools were used strictly for grammar and spelling checks**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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